Complete mappings of the Brauer and partial Brauer monoids*

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Abstract

A complete mapping of a semigroup S is a bijection $\alpha\colon S \to S$ such that $x \mapsto x \cdot x\alpha$ is also a bijection. Problem 15.11 of Araújo, Bentz, Cameron, Hendrey and Kinyon (arXiv:2608.25092) asks for a classification of complete mappings of the Brauer monoid $\mathcal{B}_n$ and the partial Brauer monoid $\mathcal{PB}_n$. By their reduction to principal factors and the Hall–Paige theorem, the open cases are the principal factors of ranks 2 and 3, together with the question of which of the monoids have a complete mapping. We settle the existence question in all cases. Every proper principal factor of $\mathcal{B}_n$ has a complete mapping. Every proper principal factor of $\mathcal{PB}_n$ has one, except the rank-2 factor of $\mathcal{PB}_3$, which is isomorphic to the Brandt semigroup $B(S_2, 3)$ and has none. Consequently $\mathcal{B}_n$ and $\mathcal{PB}_n$ have a complete mapping if and only if $n \notin \{2, 3\}$, as for the full transformation monoid and the partition monoid. The new case is that of $\mathcal{B}_n$ with $n \equiv 2, 3 \pmod 4$, in which the number of half-diagrams of rank 2 (for $n \equiv 2$) or rank 3 (for $n \equiv 3$) is odd. For it we give an explicit complete mapping, built from a triangle of half-diagrams whose sandwich permutations have odd product and from a perfect matching of the remaining half-diagrams. The other cases follow from results and methods of Araújo et al. once the Rees structure of the principal factors is written down. Every result of the paper, including the facts about diagrams and the special cases of the results of Araújo et al. that the proofs use, is formally verified in Lean 4. The proofs were found and formalised by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

Background. Let $(S, \cdot)$ be a magma. A *complete mapping* of S is a bijection $\alpha\colon S \to S$ such that the map $\theta\colon S \to S$, $x\theta = x \cdot x\alpha$, is also a bijection. As in [1], maps act on the right. For a finite magma this is the same as a transversal of the Cayley table. For finite groups, existence is decided by the Hall–Paige theorem, in the form recorded as [1, Theorem 1.1]: a finite group has a complete mapping if and only if its Sylow 2-subgroups are trivial or non-cyclic. We need only the symmetric groups: the symmetric group S_r has a complete mapping if and only if $r \notin \{2, 3\}$ (see (R1) in Section 2).

* *Version 2.* Every result of this version is formally verified in Lean 4 (Section 9); version 1 formalised only two constructions over $\mathbb{Z}_2$. A definition in Section 3 has been corrected, and four written proofs were changed (one completed, three replaced by the proofs that were formalised). Claims of version 1 for which we have no formal proof have been deleted, among them the results of the computer checks of its Section 8 and a remark comparing Lemma 4.4 with a criterion of Araújo et al.; the results of Araújo et al. that the proofs use are now stated only in the special forms that are used. The theorems are unchanged. The main changes are listed at the end of Section 9.

Araújo, Bentz, Cameron, Hendrey and Kinyon [1] developed the theory for finite semigroups. A finite semigroup has a complete mapping if and only if each of its principal factors has one [1, Theorem 4.4], which reduces the question to Rees 0-matrix semigroups; these are studied in detail in [1, Sections 6 and 7]. Among the applications, the full transformation monoid T_n and the partition monoid $\mathcal{P}_n$ have complete mappings if and only if $n = 1$ or $n \geq 4$ [1, Theorems 9.16 and 8.5], the symmetric inverse monoid I_n has one if and only if $n \equiv 0, 1 \pmod 4$ [1, Corollary 10.2], and the planar partition, Motzkin and Jones monoids have complete mappings [1, Corollary 8.8]. The Brauer monoid $\mathcal{B}_n$ and the partial Brauer monoid $\mathcal{PB}_n$ were left open. Problem 15.11 of [1] asks for a classification of their complete mappings; it notes that, by the Hall–Paige theorem, the proper principal factors still to be decided are those of ranks 2 and 3, and it also asks for which n the whole monoid has a complete mapping. Here a principal factor is *proper* if it is not the principal factor of the group of units.

Results. We decide every case.

Theorem 1.1. *Let $n \geq 1$.*

(a) Every proper principal factor of $\mathcal{B}_n$ has a complete mapping.

(b) Every proper principal factor of $\mathcal{PB}_n$ has a complete mapping, except the principal factor of rank 2 of $\mathcal{PB}_3$. That factor is isomorphic to the Brandt semigroup $B(S_2, 3) = \mathcal{M}^0(S_2, [3], [3], \Delta_3)$, where Δ_3 is the 3×3 identity matrix, and it has no complete mapping.

(c) The following are equivalent: (i) $\mathcal{B}_n$ has a complete mapping; (ii) $\mathcal{PB}_n$ has a complete mapping; (iii) S_n has a complete mapping; (iv) $n = 1$ or $n \geq 4$.

The principal factor of rank r is a Rees 0-matrix semigroup $\mathcal{M}^0(S_r, H_r, H_r, P)$ over the symmetric group S_r, indexed on both sides by the set H_r of half-diagrams of rank r (Proposition 3.1). Its sandwich matrix has identity diagonal. So when S_r has a complete mapping, that is, when $r \notin \{2, 3\}$, the factor has one by Lemma 4.1, a special case of [1, Theorem 6.2]. When $r \in \{2, 3\}$ and $N = |H_r|$ is even, it has one by Lemma 4.3, which gives an explicit form of a special case of [1, Corollary 7.6]. The remaining factors are those with $r \in \{2, 3\}$ and N odd, the case of odd index sets treated in [1, Section 7]. Table 1 lists the parities. For $\mathcal{PB}_n$ the only proper factor with $r \in \{2, 3\}$ and N odd is the rank-2 factor of $\mathcal{PB}_3$, and it has no complete mapping (Proposition 6.2). For $\mathcal{B}_n$ there are infinitely many such factors, those of rank 2 for $n \equiv 2 \pmod 4$ and of rank 3 for $n \equiv 3 \pmod 4$; the first are the rank-2 factor of $\mathcal{B}_6$, with $N = 45$, and the rank-3 factor of $\mathcal{B}_7$, with $N = 105$. For these factors we construct complete mappings explicitly (Section 5), from three pairwise compatible half-diagrams whose sandwich permutations have odd product and a perfect matching of the other half-diagrams by compatible pairs.

All our complete mappings are given by explicit formulas (Lemmas 4.1–4.4). Every result of the paper is formally verified in Lean 4 (Section 9): the facts about diagrams, including the associativity of their product (Sections 2, 3, 5 and 6), the constructions of Section 4, the special cases of the results of [1] and [4] that the proofs use, and Theorem 1.1 itself. Version 1 of this paper formalised only the two constructions of Lemmas 4.3 and 4.4 over $\mathbb{Z}_2$.

Scope. We read "classify complete mappings" in the sense explained by the second sentence of Problem 15.11: decide, for every n and every rank r, whether the principal factor of rank r has a complete mapping, and decide whether the whole monoid has one. Theorem 1.1 does this completely. We do not describe the set of all complete mappings, as [1, Proposition 6.5] does for the combinatorial Brandt semigroups by means of Latin squares.

Monoid	rank r	$N = \lvert H_r \rvert$	N odd exactly when	complete mapping of the factor
$\mathcal{B}_n$	2	$\binom{n}{2}\,(n-3)!!$	$n \equiv 2 \pmod 4$	yes: Lemma 4.3 if N is even, Section 5 if N is odd
$\mathcal{B}_n$	3	$\binom{n}{3}\,(n-4)!!$	$n \equiv 3 \pmod 4$	yes: as for rank 2
$\mathcal{PB}_n$	$2, 3$	$\binom{n}{r}\,a(n-r)$	$(n, r) = (3, 2)$	yes if N is even (Lemma 4.3); none for $(n, r) = (3, 2)$ (Proposition 6.2)

Table 1: The proper principal factors of ranks 2 and 3 ($r < n$, and $r \equiv n \pmod 2$ for $\mathcal{B}_n$). Here N is the number of half-diagrams of rank r (Section 3) and $a(k)$ is the number of involutions of a k-set. The factors of rank $r \notin \{2, 3\}$ have complete mappings by Lemma 4.1.

Prior work. Problem 15.11 appears in version 1 of [1], submitted to the arXiv on 25 August 2026, which was still the only version when we checked on 26 September 2026. We found no solution of, and no announced progress on, Problem 15.11 in the following sources, which we searched on 26 September 2026 between 06:49 and 07:46 UTC: the arXiv (the listing of [1] and eight title and abstract searches through the arXiv API); GitHub searches of issues, pull requests, commits, code and repositories; the public repositories of AI-assisted work on open problems that we check, namely the file tree, issues and pull requests of the `trureturing` repository [8], and the repositories SCOPE2026, AI-Has-Taste, formal-conjectures and astrafala; the problem database MathDB (`https://mathdb.com`); MathOverflow and Mathematics Stack Exchange, through their search interface; Crossref; web search; the blog post announcing [1] and its comments [2], and the publication list and arXiv listing of the third author; and the slides of a 2025 talk on the subject by the fifth author [6]. The only earlier AI-assisted work on [1] that we found is issue #9377 and pull request #9405 of [8], which concern Problem 15.5 and do not mention Problem 15.11 or diagram monoids. Our searches on 26 September 2026 for works citing [1] in OpenAlex and Semantic Scholar failed (HTTP error 429, rate limit exceeded), so citing works were not checked systematically; we did not search Google Scholar, zbMATH or MathSciNet. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, where [1] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. Before the release of version 2 we repeated the searches on 6 October 2026 between 17:25 and 19:18 UTC: the arXiv, where [1] still had only version 1, including all new submissions and replacements in math.GR, math.RA and math.CO announced up to 6 October 2026; works citing [1] in Semantic Scholar, OpenAlex and DataCite, which list none; zbMATH Open, whose search interface was unavailable, through its OAI-PMH interface, where no record in the Mathematics Subject Classification classes 05 and 20 added or changed since 25 August 2026 cites [1]; the blog post [2], which still had no comments, and the web pages of the authors; MathDB; MathOverflow and Mathematics Stack Exchange; GitHub, including [8] and other repositories of AI-assisted work on open problems; and web search. We again found no solution of Problem 15.11 and no announced progress on it; we again did not search Google Scholar or MathSciNet. A last check on 7 October 2026 between 01:31 and 01:34 UTC, of the arXiv (where [1] still had only version 1, including the new submissions in math.GR, math.RA and math.CO announced that day) and of GitHub, including [8], found nothing new. This reports the coverage of our searches, not a guarantee of priority. Our results are a direct extension of the methods of [1]: the reduction to principal factors, the criteria for Rees 0-matrix semigroups and the lifting lemma, of which we use special cases, are theirs; version 1 of this paper derived Proposition 6.2 from their non-existence theorem [1, Theorem 7.19], for which version 2 substitutes a direct parity argument; and our treatment of $\mathcal{B}_n$ follows their proof for the partition monoid [1, Theorem 8.5], including the triangle of half-diagrams in Lemma 5.2.

Organisation. Section 2 fixes notation and lists the results of [1] that we use, in the special forms in which we use them. Section 3 gives the Rees structure of the principal factors of $\mathcal{B}_n$ and $\mathcal{PB}_n$. Section 4 gives explicit complete mappings of Rees 0-matrix semigroups with identity diagonal. Section 5 treats $\mathcal{B}_n$ and proves Theorem 1.1(a), Section 6 treats $\mathcal{PB}_n$ and proves Theorem 1.1(b), and Section 7 proves Theorem 1.1(c). Section 8 describes the computer checks of version 1, and Section 9 the formal verification.

2 Preliminaries

Rees 0-matrix semigroups. We follow [1, Section 4]. Let G be a finite group, let I and Λ be non-empty finite sets, and let $P = (p_{\lambda i})$ be a $\Lambda \times I$ matrix over $G \cup \{0\}$ with no zero row or column. The Rees 0-matrix semigroup $\mathcal{M}^0(G, I, \Lambda, P)$ is $(I \times G \times \Lambda) \cup \{0\}$ with the product

$$(i, a, \lambda)(j, b, \mu) = \begin{cases} (i,\ a\, p_{\lambda j}\, b,\ \mu) & \text{if } p_{\lambda j} \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and $0x = x0 = 0$. For a $\mathcal{J}$-class J of a finite semigroup S, the principal factor J^0 is $J \cup \{0\}$, with the product of S when it lies in J and 0 otherwise.

Results that we use. We use the following special cases of results of [1], with the numbering of [1]. Each of them is proved in our Lean formalisation (Section 9): (R1) and (R2) as the statements `SymCM` and `ZeroFixed`, (R3) and (R4) as lemmas inside the proof of Theorem 1.1(c). The general results are not formalised.

(R1) The symmetric group S_r has a complete mapping if and only if $r \notin \{2, 3\}$. This is the case of the symmetric groups of the Hall–Paige theorem [1, Theorem 1.1]; for the symmetric groups it is due to Hall and Paige [5, Theorems 2 and 5].

(R2) [1, Proposition 1.4]. A complete mapping of a magma with zero fixes the zero; hence so does the corresponding map θ, since $0 \cdot 0 = 0$.

(R3) [1, Corollary 3.4], for $\mathcal{B}_n$ and $\mathcal{PB}_n$. If $\mathcal{B}_n$ or $\mathcal{PB}_n$ has a complete mapping, then so does its group of units.

(R4) [1, Theorem 4.4], the "if" direction, for $\mathcal{B}_n$ and $\mathcal{PB}_n$. If every principal factor J^0 of $\mathcal{B}_n$, respectively $\mathcal{PB}_n$, has a complete mapping, then so does $\mathcal{B}_n$, respectively $\mathcal{PB}_n$.

Diagrams. Let $[n] = \{1, \ldots, n\}$ and $[n]' = \{1', \ldots, n'\}$. The partial Brauer monoid $\mathcal{PB}_n$ consists of the partitions of $[n] \cup [n]'$ into blocks of size 1 or 2, and the Brauer monoid $\mathcal{B}_n$ of those with all blocks of size 2; see [4, Section 2.2]. We draw $[n]$ as the top row and $[n]'$ as the bottom row. The product xy is formed by drawing x above y, identifying the bottom row of x with the top row of y (the *middle row*), keeping the connectivity of the outer points, and discarding closed loops and components contained in the middle row. The product is associative, and the diagram whose blocks are the sets $\{i, i'\}$ is an identity for it, so $\mathcal{PB}_n$ and $\mathcal{B}_n$ are monoids. A block meeting both rows is a *through block*; the *rank* of x is its number of through blocks, and $\operatorname{rank}(xy) \leq \min(\operatorname{rank} x, \operatorname{rank} y)$. As recalled in [1, Section 15], it is proved in [4, Proposition 2.1 and Remark 2.2(v)–(vii)] that the $\mathcal{J}$-classes of $\mathcal{B}_n$ and $\mathcal{PB}_n$ are the rank classes. We write J_r for the set of elements of rank r; in $\mathcal{B}_n$ it is non-empty exactly when $r \leq n$ and $r \equiv n \pmod 2$. The group of units is J_n, the set of permutation diagrams, isomorphic to S_n.

3 Rees coordinates

Let $D \in \{\mathcal{B}_n, \mathcal{PB}_n\}$ and let r be a rank that occurs in D. A *half-diagram* of rank r is a pair $h = (F, M)$, where $F = F(h) \subseteq [n]$ has $|F| = r$ (the *free points*) and $M = M(h)$ is a set of disjoint 2-subsets (*arcs*) of $[n] \setminus F$ covering $[n] \setminus F$ if $D = \mathcal{B}_n$, and arbitrary if $D = \mathcal{PB}_n$; the points of $[n] \setminus F$ not covered by M are *singletons*. Let $H = H_r(D)$ be the set of half-diagrams of rank r and $N = |H|$. Then

$$N = \binom{n}{r}(n-r-1)!! \text{ for } \mathcal{B}_n, \qquad N = \binom{n}{r} a(n-r) \text{ for } \mathcal{PB}_n,$$

where $(-1)!! = 1$ and $a(k)$ is the number of involutions of a k-set, that is, of permutations σ with $\sigma^2 = 1$, the identity included.

For $x \in J_r$, let $\mathrm{top}(x) = (F, M)$, where F is the set of top points of through blocks and M the set of blocks of size 2 contained in the top row (the blocks of size 1 in the top row are the singletons); let $\mathrm{bot}(x)$ be defined in the same way from the bottom row, with primes removed. If $a_1 < \cdots < a_r$ are the top points and $b_1 < \cdots < b_r$ the bottom points (without primes) of the through blocks, define $g(x) \in S_r$ by requiring that $\{a_k, b'_{k\,g(x)}\}$ is a block of x for every k. Permutations act on the right, and gh means first g, then h.

For $\lambda, i \in H$, let $\Gamma(\lambda, i)$ be the multigraph on $[n]$ whose edges are the arcs of λ and the arcs of i (an arc of both gives a double edge). Every vertex has degree at most 2, so the components are paths and cycles. A point of $F(\lambda)$ lies on no arc of λ, so it has degree at most 1 and is an end of a path. We say that λ and i are *compatible*, and write $\lambda \sim i$, if the path starting at each point of $F(\lambda)$ ends at a point of $F(i)$; a point of $F(\lambda) \cap F(i)$ is a path of length 0. The r paths are distinct components (a point of $F(\lambda) \cap F(i)$ is isolated, so no path joins two points of $F(\lambda)$ when $\lambda \sim i$), so they then match $F(\lambda)$ bijectively with $F(i)$, and we define $p_{\lambda i} \in S_r$ by requiring that the k-th smallest point of $F(\lambda)$ is joined to the $(k\,p_{\lambda i})$-th smallest point of $F(i)$.

Proposition 3.1. *Let $D \in \{\mathcal{B}_n, \mathcal{PB}_n\}$, let r be a rank that occurs in D, and let $H = H_r(D)$.*

(a) The map $x \mapsto (\mathrm{top}(x), g(x), \mathrm{bot}(x))$ is a bijection from J_r to $H \times S_r \times H$.

(b) For $x, y \in J_r$ we have $\mathrm{rank}(xy) = r$ if and only if $\mathrm{bot}(x) \sim \mathrm{top}(y)$, and then $\mathrm{top}(xy) = \mathrm{top}(x)$, $\mathrm{bot}(xy) = \mathrm{bot}(y)$ and $g(xy) = g(x)\, p_{\mathrm{bot}(x),\mathrm{top}(y)}\, g(y)$.

(c) For all $\lambda, i \in H$: $\lambda \sim \lambda$ and $p_{\lambda\lambda} = 1$; and $\lambda \sim i$ if and only if $i \sim \lambda$, in which case $p_{i\lambda} = p_{\lambda i}^{-1}$.

(d) If $F(\lambda) = F(i)$, then $\lambda \sim i$ and $p_{\lambda i} = 1$.

Consequently $J_r^0 \cong \mathcal{M}^0(S_r, H, H, P)$, where P is the $H \times H$ matrix with entries $p_{\lambda i}$ if $\lambda \sim i$ and 0 otherwise; the isomorphism extends the map in (a) by $0 \mapsto 0$.

Proof. (a) An element of rank r is determined by its blocks inside the top row, its blocks inside the bottom row, and a bijection from its top through points to its bottom through points, and every such choice occurs. The first two data are $\mathrm{top}(x)$ and $\mathrm{bot}(x)$, and $g(x)$ encodes the bijection.

(b) The blocks of x inside the top row are blocks of xy. Consider the top point a_k of a through block of x. Its strand reaches the middle row at $b = b_{k\,g(x)} \in F(\mathrm{bot}(x))$ and continues along the path of $\Gamma(\mathrm{bot}(x), \mathrm{top}(y))$ starting at b. This path alternates between arcs of $\mathrm{top}(y)$ and arcs of $\mathrm{bot}(x)$, beginning with an arc of $\mathrm{top}(y)$, and stops at a point e at which the next arc is missing. So either $e \in F(\mathrm{top}(y))$, and the strand continues through y to the bottom row; or $e \neq b$ and $e \in F(\mathrm{bot}(x))$, and a_k is joined to another top through point of x; or e is a singleton of $\mathrm{bot}(x)$ or of $\mathrm{top}(y)$ (only in $\mathcal{PB}_n$), and a_k becomes a singleton. Every through block of xy contains a top point of a through block of x, so $\mathrm{rank}(xy) = r$ if and only if the first case occurs

for all k, that is, if and only if $\mathrm{bot}(x) \sim \mathrm{top}(y)$. In that case the blocks of xy inside the top row are exactly those of x, so $\mathrm{top}(xy) = \mathrm{top}(x)$. The r strands end at r distinct bottom through points of y, that is, at all of them, so the blocks of xy inside the bottom row are those of y and $\mathrm{bot}(xy) = \mathrm{bot}(y)$. Following a_k: it reaches the $(k\,g(x))$-th smallest point of $F(\mathrm{bot}(x))$, which is joined to the $(k\,g(x)\,p)$-th smallest point of $F(\mathrm{top}(y))$, where $p = p_{\mathrm{bot}(x),\mathrm{top}(y)}$, and this point is joined in y to the $(k\,g(x)\,p\,g(y))$-th smallest bottom through point of y.

(c) In $\Gamma(\lambda,\lambda)$ each arc of λ is a double edge and each point of $F(\lambda)$ is isolated, which gives $\lambda \sim \lambda$ and $p_{\lambda\lambda} = 1$. The graphs $\Gamma(\lambda,i)$ and $\Gamma(i,\lambda)$ coincide. If $\lambda \sim i$, the r paths starting in $F(\lambda)$ end at r distinct points of $F(i)$, hence at all of them; so the path starting at a point of $F(i)$ is one of these paths, read backwards. Hence $i \sim \lambda$ and $p_{i\lambda} = p_{\lambda i}^{-1}$.

(d) Each point of $F(\lambda) = F(i)$ lies on no arc of λ or of i, so it is a path of length 0.

Finally, J_r is a $\mathcal{J}$-class (Section 2), and $\mathrm{rank}(xy) \le r$ for $x, y \in J_r$. So by (b) the product of $x, y \in J_r$ in J_r^0 is xy if $\mathrm{bot}(x) \sim \mathrm{top}(y)$ and 0 otherwise, and by (a) and (b) the stated map is an isomorphism. By (c), P has no zero row or column. □

The rows of P are indexed by bottom halves and the columns by top halves. Proposition 3.1 is the analogue for $\mathcal{B}_n$ and $\mathcal{PB}_n$ of [1, Proposition 8.3] for the partition monoid.

For $\lambda \in H$, let $e_\lambda \in J_r$ be the element with coordinates $(\lambda, 1, \lambda)$, that is, the diagram whose top and bottom halves are both λ and whose through blocks are the pairs $\{a, a'\}$ with $a \in F(\lambda)$. Since $p_{\lambda\lambda} = 1$, Proposition 3.1(b) gives $(\lambda, g, \lambda)(\lambda, h, \lambda) = (\lambda, gh, \lambda)$ for $g, h \in S_r$. So e_λ is an idempotent, and the set $\{(\lambda, g, \lambda) : g \in S_r\}$ is a group with identity e_λ, isomorphic to S_r by $x \mapsto g(x)$.

4 Explicit complete mappings

Throughout this section H is a non-empty finite set, $r \ge 0$, and P is an $H \times H$ matrix over $S_r \cup \{0\}$ with $p_{\lambda\lambda} = 1$ for all λ. We write $\lambda \sim \mu$ when $p_{\lambda\mu} \ne 0$. By Proposition 3.1, the principal factors of $\mathcal{B}_n$ and $\mathcal{PB}_n$ are of this form. We fix a Latin square on H, that is, a map $L\colon H \times H \to H$ such that $L(i,\cdot)$ and $L(\cdot,\lambda)$ are bijections for all i, λ; for instance, identify H with $\mathbb{Z}_N$ and put $L(i,\lambda) = i + \lambda$.

Lemma 4.1. *If S_r has a complete mapping φ, then the map f with $0f = 0$ and $(i,g,\lambda)f = (\lambda,\ g\varphi,\ L(i,\lambda))$ is a complete mapping of $\mathcal{M}^0(S_r, H, H, P)$.*

Proof. The map $(i,\lambda) \mapsto (\lambda, L(i,\lambda))$ is a bijection of $H \times H$ and φ is a bijection, so f is a bijection. Since $p_{\lambda\lambda} = 1$, we get $(i,g,\lambda)\cdot(i,g,\lambda)f = (i,\ g \cdot g\varphi,\ L(i,\lambda)) \ne 0$, and this is a bijection because $(i,\lambda) \mapsto (i, L(i,\lambda))$ and $g \mapsto g \cdot g\varphi$ are bijections. □

Lemma 4.1 uses only the diagonal of P, as in [1, Proposition 6.5]; its existence statement is a special case of [1, Theorem 6.2].

Now let $r \in \{2, 3\}$. Let $s\colon S_r \to \mathbb{Z}_2 = \{0, 1\}$ be the sign homomorphism written additively, with $s(g) = 0$ exactly for even g; its kernel is the alternating group A_r. We write $\mathcal{M}_2$ for the Rees 0-matrix semigroup $\mathcal{M}^0(\mathbb{Z}_2, H, H, s(P))$, with $\mathbb{Z}_2$ written additively: its non-zero elements are the triples $(i,a,\lambda) \in H \times \mathbb{Z}_2 \times H$, and

$$(i,a,\lambda)(j,b,\mu) = (i,\ a + s(p_{\lambda j}) + b,\ \mu) \ \text{ if } \lambda \sim j, \qquad (i,a,\lambda)(j,b,\mu) = 0 \ \text{ otherwise.}$$

Note that $s(p_{\lambda\lambda}) = 0$.

Lemma 4.2 (lifting)**.** *Let $r \in \{2,3\}$, and suppose that $\mathcal{M}_2$ has a complete mapping. Then $\mathcal{M}^0(S_r, H, H, P)$ also has a complete mapping.*

Proof. We follow the proof of [1, Theorem 5.2], which [1, Lemma 6.3] extends to Rees 0-matrix semigroups, in the case of the normal subgroup A_r of S_r. The group A_r has odd order (1 or 3), so $h \mapsto h^2$ is a bijection of A_r. Put $g_0 = 1$ and $g_1 = (1\,2)$, and write each $g \in S_r$ as $g = g_u h$ with $u = s(g)$ and $h \in A_r$. Let $\bar{f}$ be a complete mapping of $\mathcal{M}_2$. By (R2), $\bar{f}$ and the map $x \mapsto x \cdot x\bar{f}$ fix 0; as they are bijections, the product of a non-zero x with $x\bar{f}$ is non-zero. So if $(i,u,\lambda)\bar{f} = (j,v,\mu)$, then $\lambda \sim j$, and we may put $(i, g_u h, \lambda)f = (j,\, p^{-1}hp\, g_v,\, \mu)$ with $p = p_{\lambda j}$, and $0f = 0$. Since $h \mapsto p^{-1}hp\, g_v$ is a bijection from A_r onto the set of elements of sign v, and $\bar{f}$ is a bijection, f is a bijection. Moreover $(i, g_u h, \lambda) \cdot (i, g_u h, \lambda)f = (i,\, g_u h^2 p\, g_v,\, \mu)$; for fixed (i,u,λ) this runs bijectively through the elements of sign $u + s(p) + v$, which is the middle coordinate of the product $(i,u,\lambda) \cdot (i,u,\lambda)\bar{f}$ in $\mathcal{M}_2$. Since $x \mapsto x \cdot x\bar{f}$ is a bijection of $\mathcal{M}_2$, the map $x \mapsto x \cdot xf$ is a bijection. □

Lemma 4.3. *Let $r \in \{2,3\}$. Let δ be a permutation of H and $t\colon H \to \mathbb{Z}_2$ a map with $t(\mu\delta) = t(\mu) + 1$ for all $\mu \in H$. Define $\bar{f}$ on $\mathcal{M}_2$ by $0\bar{f} = 0$ and, writing $\mu = L(i,\lambda)$,*

$$(i,\, 1 + t(\mu),\, \lambda)\bar{f} = (\lambda,\, 1,\, \mu), \qquad (i,\, t(\mu),\, \lambda)\bar{f} = (\lambda,\, 0,\, \mu\delta).$$

Then $\bar{f}$ is a complete mapping of $\mathcal{M}_2$. Such δ and t exist when $N = |H|$ is even, so in that case $\mathcal{M}^0(S_r, H, H, P)$ has a complete mapping.

Proof. The element $(\lambda, 1, \nu)$ is the image of exactly one element, coming from the unique i with $L(i,\lambda) = \nu$; the element $(\lambda, 0, \nu)$ is the image of exactly one element, coming from the unique i with $L(i,\lambda)\delta = \nu$. So $\bar{f}$ is a bijection. Since $s(p_{\lambda\lambda}) = 0$, the products are

$$(i, 1{+}t(\mu), \lambda)(\lambda, 1, \mu) = (i,\, t(\mu),\, \mu), \qquad (i, t(\mu), \lambda)(\lambda, 0, \mu\delta) = (i,\, t(\mu),\, \mu\delta) = (i,\, t(\mu\delta){+}1,\, \mu\delta).$$

For fixed i, $\mu = L(i,\lambda)$ runs through H as λ does, so the products are the elements $(i, t(\nu), \nu)$ and $(i, t(\nu) + 1, \nu)$ for $\nu \in H$, each exactly once. If N is even, let δ be a fixed-point-free involution of H and let t take the values 0 and 1 on the two points of each 2-cycle. The last statement follows from Lemma 4.2. □

The existence statement of Lemma 4.3 is a special case of [1, Corollary 7.6]; the lemma gives an explicit formula.

Lemma 4.4. *Let $r \in \{2,3\}$. Let ρ be a permutation of H with $\lambda \sim \lambda\rho$ for all $\lambda \in H$ such that, for every cycle $(\lambda_0, \lambda_1, \ldots, \lambda_{m-1})$ of ρ, with indices modulo m,*

$$m + \sum_{t=0}^{m-1} s(p_{\lambda_t \lambda_{t+1}}) \equiv 0 \pmod 2. \tag{$*$}$$

Then there is a map $c\colon H \to \mathbb{Z}_2$ with $c(\lambda\rho) = c(\lambda) + 1 + s(p_{\lambda,\lambda\rho})$ for all λ. For such c, the map $\bar{f}$ with $0\bar{f} = 0$ and

$$(i,\, c(\lambda),\, \lambda)\bar{f} = (\lambda,\, 1 + c(\lambda),\, L(i,\lambda)), \qquad (i,\, 1 + c(\lambda),\, \lambda)\bar{f} = (\lambda\rho,\, c(\lambda\rho),\, L(i,\lambda\rho))$$

is a complete mapping of $\mathcal{M}_2$, and hence $\mathcal{M}^0(S_r, H, H, P)$ has a complete mapping.

Proof. On each cycle of ρ, choose $c(\lambda_0)$ arbitrarily and define $c(\lambda_{t+1}) = c(\lambda_t) + 1 + s(p_{\lambda_t\lambda_{t+1}})$ for $t = 0, \ldots, m-2$; condition ($*$) says exactly that the required relation also holds for $t = m-1$. The images under the first rule are the elements $(\lambda, 1 + c(\lambda), \mu)$, and those under the second rule the elements $(\kappa, c(\kappa), \mu)$ with $\kappa = \lambda\rho$; in both cases μ runs through H as i does. These two sets are complementary, so $\bar{f}$ is a bijection. The products are

$$(i,\, c(\lambda) + 0 + 1 + c(\lambda),\, L(i,\lambda)) = (i, 1, L(i,\lambda))$$

from the first rule, which is non-zero since $\lambda \sim \lambda$, and

$$(i,\ 1 + c(\lambda) + s(p_{\lambda,\lambda\rho}) + c(\lambda\rho),\ L(i,\lambda\rho)) = (i, 0, L(i,\lambda\rho))$$

from the second, which is non-zero since $\lambda \sim \lambda\rho$. For fixed i they fill $\{i\} \times \{1\} \times H$ and $\{i\} \times \{0\} \times H$ bijectively. The last statement follows from Lemma 4.2. □

Lemma 4.4 is an explicit instance of the methods of [1, Section 7].

Remark 4.5. (1) A fixed point of ρ would be a cycle with $m = 1$ and sum 0, so ρ has no fixed points. (2) If $\lambda \sim \mu$ implies $\mu \sim \lambda$ and $p_{\mu\lambda} = p_{\lambda\mu}^{-1}$, as in Proposition 3.1(c), then every transposition $(a\ b)$ with $a \sim b$ satisfies $(*)$, because $s(p_{ab}) = s(p_{ba})$. (3) A 3-cycle $(a\ b\ c)$ satisfies $(*)$ if and only if $s(p_{ab}) + s(p_{bc}) + s(p_{ca}) = 1$, that is, if and only if $p_{ab}p_{bc}p_{ca}$ is odd.

5 The Brauer monoid

In this section $D = \mathcal{B}_n$, $r \in \{2, 3\}$, $r < n$ and $n \equiv r \pmod 2$, and $H = H_r(\mathcal{B}_n)$. For an r-subset F of $[n]$, the *fibre* over F is $H_F = \{h \in H : F(h) = F\}$. It has $m = (n - r - 1)!!$ elements, an odd number, and by Proposition 3.1(d) any two of its elements are compatible.

Lemma 5.1. *$N = |H|$ is odd if and only if $r = 2$ and $n \equiv 2 \pmod 4$, or $r = 3$ and $n \equiv 3 \pmod 4$. In these cases $n - r \geq 4$ and $\binom{n}{r}$ is odd.*

Proof. Since m is odd, $N = \binom{n}{r}m$ is odd if and only if $\binom{n}{r}$ is odd. By Lucas's theorem for the prime 2, in the form $\binom{n}{k} \equiv \binom{n \bmod 2}{k \bmod 2}\binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor} \pmod 2$, we have $\binom{n}{2} \equiv \lfloor n/2 \rfloor$ and $\binom{n}{3} \equiv (n \bmod 2)\lfloor n/2 \rfloor \pmod 2$. So for even n, $\binom{n}{2}$ is odd if and only if $n \equiv 2 \pmod 4$, and for odd n, $\binom{n}{3}$ is odd if and only if $n \equiv 3 \pmod 4$. Since $r < n$, this gives $n \geq 6$ for $r = 2$ and $n \geq 7$ for $r = 3$, so $n - r \geq 4$. □

Lemma 5.2 (the triangle)**.** *Suppose that $n - r \geq 2$. Let $E = \varnothing$ if $r = 2$ and $E = \{5\}$ if $r = 3$, and let R_0 be the perfect matching $\{r+3, r+4\}, \{r+5, r+6\}, \ldots$ of $\{r+3, \ldots, n\}$. Put*

$$\begin{aligned} I_0 &= (\{3,4\} \cup E,\ \{\{1,2\}\} \cup R_0), \\ I_1 &= (\{2,4\} \cup E,\ \{\{1,3\}\} \cup R_0), \\ I_2 &= (\{2,3\} \cup E,\ \{\{1,4\}\} \cup R_0). \end{aligned}$$

Then $I_0 \sim I_1$, $I_1 \sim I_2$ and $I_2 \sim I_0$, with $p_{I_0I_1} = p_{I_1I_2} = 1$ and $p_{I_2I_0} = (1\,2)$. In particular $p_{I_0I_1}p_{I_1I_2}p_{I_2I_0}$ is odd.

Proof. The arcs of R_0 are common to the three half-diagrams and give double edges, and the point 5 (if $r = 3$) is free in all three and is the largest free point, so it is joined to itself. Only the points $1, \ldots, 4$ matter. In $\Gamma(I_0, I_1)$ the path from 3 is $3, 1, 2$ (arcs $\{1,3\}$ of I_1 and $\{1,2\}$ of I_0) and ends at $2 \in F(I_1)$, and 4 is fixed; the order is preserved. In $\Gamma(I_1, I_2)$, 2 is fixed and the path from 4 is $4, 1, 3$; the order is preserved. In $\Gamma(I_2, I_0)$ the path from 2 is $2, 1, 4$ and 3 is fixed, so the smaller free point 2 of I_2 is joined to the larger free point 4 of I_0, and $p_{I_2I_0} = (1\,2)$. □

These are the three half-diagrams used for $\mathcal{P}_n$ in the proof of [1, Theorem 8.5], where the marked singletons become free points, completed to perfect matchings by R_0. With the idempotents e_λ defined after Proposition 3.1, Lemma 5.2 says that $e_{I_0}e_{I_1}e_{I_2}e_{I_0} = (I_0,\ p_{I_0I_1}p_{I_1I_2}p_{I_2I_0},\ I_0)$ is an odd element of the group $\{(I_0, g, I_0) : g \in S_r\}$.

Lemma 5.3 (joining two fibres)**.** *Let $F \neq F'$ be r-subsets of $[n]$ with $F \setminus F' = \{x_1, \ldots, x_k\}$ and $F' \setminus F = \{y_1, \ldots, y_k\}$, and suppose that $r + 2k \leq n$. Then there are $v \in H_F$ and $w \in H_{F'}$ with $v \sim w$.*

Proof. Choose distinct points $z_1, \ldots, z_k \notin F \cup F'$, which is possible since $|F \cup F'| + k = r + 2k \leq n$, and a perfect matching R of the remaining $n - r - 2k$ points, an even number. Put

$$v = (F,\ \{\{z_t, y_t\} : t \leq k\} \cup R), \qquad w = (F',\ \{\{x_t, z_t\} : t \leq k\} \cup R).$$

In $\Gamma(v, w)$ the path from x_t is x_t, z_t, y_t and ends in F', the points of $F \cap F'$ are paths of length 0, and the arcs of R give double edges. So $v \sim w$. □

Lemma 5.4 (matching)**.** *Suppose that $n - r \geq 4$ and $\binom{n}{r}$ is odd, and let $X = \{I_0, I_1, I_2\}$ be as in Lemma 5.2. Then $H \setminus X$ can be partitioned into pairs $\{v, w\}$ with $v \sim w$.*

Proof. The elements of X lie in the three distinct fibres over $\{3,4\} \cup E$, $\{2,4\} \cup E$ and $\{2,3\} \cup E$, one in each. The other $\binom{n}{r} - 3$ fibres, an even number, can be paired so that each pair (F, F') satisfies $r + 2|F \setminus F'| \leq n$. Indeed, if $r = 2$ then $|F \setminus F'| \leq 2$ and $n \geq 6$, and if $r = 3$ and $n \geq 9$ then $|F \setminus F'| \leq 3$; in these cases any pairing works. If $(n, r) = (7, 3)$ the condition is $F \cap F' \neq \varnothing$, and the following 16 pairs of 3-subsets of $[7]$ work: $\{a, b, 6\}$ with $\{a, b, 7\}$ for the ten 2-subsets $\{a, b\}$ of $[5]$; $\{1, 6, 7\}$ with $\{1, 2, 3\}$, $\{2, 6, 7\}$ with $\{1, 2, 4\}$, $\{3, 6, 7\}$ with $\{1, 3, 4\}$, $\{4, 6, 7\}$ with $\{2, 3, 4\}$ and $\{5, 6, 7\}$ with $\{1, 2, 5\}$; and $\{1, 3, 5\}$ with $\{1, 4, 5\}$. These are exactly the 32 subsets other than $\{3, 4, 5\}$, $\{2, 4, 5\}$ and $\{2, 3, 5\}$, and the two sets in each pair meet. For each pair of fibres, Lemma 5.3 gives compatible $v \in H_F$ and $w \in H_{F'}$, and we take $\{v, w\}$ as a pair. Now exactly one element of each fibre has been used, either in X or in a pair. The remaining $m - 1$ elements of each fibre, an even number, are pairwise compatible, and we pair them arbitrarily. □

Proof of Theorem 1.1(a). Let $r < n$ be a rank that occurs in $\mathcal{B}_n$. By Proposition 3.1, $J_r^0 \cong \mathcal{M}^0(S_r, H, H, P)$ with $p_{\lambda\lambda} = 1$ for all λ. If $r \notin \{2, 3\}$, then S_r has a complete mapping by (R1), and Lemma 4.1 applies. If $r \in \{2, 3\}$ and N is even, Lemma 4.3 applies. If $r \in \{2, 3\}$ and N is odd, then $n - r \geq 4$ and $\binom{n}{r}$ is odd by Lemma 5.1. Let ρ be the permutation of H that maps $I_0 \mapsto I_1 \mapsto I_2 \mapsto I_0$ and interchanges the two elements of each pair given by Lemma 5.4. Then $\lambda \sim \lambda\rho$ for all λ. The 3-cycle satisfies $(*)$ by Lemma 5.2 and Remark 4.5(3), and the transpositions satisfy it by Proposition 3.1(c) and Remark 4.5(2). So Lemma 4.4 gives a complete mapping of J_r^0. □

6 The partial Brauer monoid

Lemma 6.1. *The number $a(k)$ of involutions of a k-set is even for $k \geq 2$. Hence, if $n \geq 4$, $r \in \{2, 3\}$ and $r < n$, then $N = \binom{n}{r} a(n - r)$ is even.*

Proof. The map $\sigma \mapsto \sigma^{-1}$ is an involution of the symmetric group on a k-set. Its fixed points are the $a(k)$ permutations with $\sigma^2 = 1$, and its other orbits have two elements. So $a(k) \equiv k!$ (mod 2), which is even for $k \geq 2$. If $n - r \geq 2$ this gives the claim; otherwise $n - r = 1$, so $(n, r) = (4, 3)$ and $N = 4$. □

Proposition 6.2. *The principal factor of rank 2 of $\mathcal{PB}_3$ is isomorphic to the Brandt semigroup $B(S_2, 3) = \mathcal{M}^0(S_2, [3], [3], \Delta_3)$ and has no complete mapping.*

Proof. The half-diagrams of rank 2 are the three pairs $(F, \varnothing)$ with $|F| = 2$, the third point being a singleton. If $F(\lambda) \neq F(i)$, a point of $F(\lambda) \setminus F(i)$ is a singleton of i, hence an isolated vertex of $\Gamma(\lambda, i)$ outside $F(i)$, so $\lambda \not\sim i$. With Proposition 3.1(c), the sandwich matrix is the identity matrix Δ_3 over S_2, and Proposition 3.1 gives the isomorphism.

Let $M = \mathcal{M}^0(S_2, [3], [3], \Delta_3)$, suppose that α is a complete mapping of M, and let $x\theta = x \cdot x\alpha$. By (R2), α and θ fix 0, so they permute the 18 non-zero elements of M. For a non-zero element $x = (i, g, \lambda)$ put $\sigma(x) = s(g) \in \mathbb{Z}_2$, where s is the sign. Exactly 9 of the non-zero elements have

$\sigma(x) = 1$, so $\sum_{x \neq 0} \sigma(x) = 1$ in $\mathbb{Z}_2$. For $x \neq 0$ the product $x \cdot x\alpha$ is non-zero, and since the non-zero entries of Δ_3 are the identity, $\sigma(x\theta) = \sigma(x) + \sigma(x\alpha)$. Summing over the non-zero elements and using that θ and α permute them, we get $1 = \sum \sigma(x\theta) = \sum \sigma(x) + \sum \sigma(x\alpha) = 1 + 1 = 0$ in $\mathbb{Z}_2$, a contradiction. □

Proof of Theorem 1.1(b). Let $r < n$. If $r \notin \{2, 3\}$, Lemma 4.1 applies as in the proof of part (a). If $r \in \{2, 3\}$ and $n \geq 4$, then N is even by Lemma 6.1, and Lemma 4.3 applies. If $r \in \{2, 3\}$ and $n \leq 3$, then $(n, r) = (3, 2)$, and Proposition 6.2 applies. □

7 The whole monoids

Proof of Theorem 1.1(c). The group of units of $\mathcal{B}_n$ and of $\mathcal{PB}_n$ is $J_n \cong S_n$, so (i) and (ii) each imply (iii) by (R3). Statements (iii) and (iv) are equivalent by (R1). Suppose that (iii) holds, and let φ be a complete mapping of S_n. The principal factor of the group of units is S_n with a zero adjoined, and φ extended by $0 \mapsto 0$ is a complete mapping of it. Every proper principal factor of $\mathcal{B}_n$ has a complete mapping by part (a). Every proper principal factor of $\mathcal{PB}_n$ has one by part (b), since $n \neq 3$. So (i) and (ii) hold by (R4). □

For $n = 0$ the monoids $\mathcal{B}_0$ and $\mathcal{PB}_0$ and the group S_0 have one element each, and each has a complete mapping. So (i), (ii) and (iii) are equivalent to $n \notin \{2, 3\}$ for every $n \geq 0$; this is the form stated in the abstract.

8 Computations

In version 2 every result of this paper is proved in Lean (Section 9), and no statement depends on a computation. Version 1 reported computer checks of its proofs and of our conventions: two implementations of the product of diagrams that share no code, certificates of complete mappings of some principal factors and of $\mathcal{B}_n$ and $\mathcal{PB}_n$ for small n, and an exhaustive search for Proposition 6.2. They are not part of version 2, and we make no claim about them here; the programs, certificates and logs remain in the directory `certificates/brauer-complete` of version 1.4.0 of the repository [3].

9 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [7] (version v4.34.1, commit `d13f23b7`). Version 1 of this paper was released with a partial formalisation, the module `BrauerCMCore.lean` of version 1.4.0 of [3], which proved that the maps $\bar{f}$ of Lemmas 4.3 and 4.4 are complete mappings of $\mathcal{M}_2$, for every finite index set and every matrix over $\mathbb{Z}_2$ with zero diagonal. For version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 2, a definition, a step of a written proof, or an observation that is immediate from the definitions.

Statements before formal proofs. In short: the statements were written from the released paper, reviewed independently and fixed by their hash before the new formal proofs were written, and the proofs import the statement file without modifying it. In detail, the statements were written after the written proofs: they formalise a paper that had already been released. On 27 September 2026, after version 1 had been released, an agent wrote a Lean file of definitions and statements, the *statement file*, which proves none of its statements, and a file of 68 known-answer tests of its definitions. An independent agent reviewed the statement file against the paper on 27 September 2026, between 18:18 and 18:56 UTC, checked it on examples of its own and recommended freezing it as it was; its SHA-256 hash is `ca2ff711b3ed008112d3633551a0703d`

a79167805815d0cbe04599728a9bc373, and it has not changed since. The proofs were written in separate files that import the statement file and never modify it. The earliest of them that was kept was created on 29 September 2026 at 21:43 UTC, and the first group of proofs was submitted for acceptance at 21:49 UTC. The proofs of Lemmas 4.3 and 4.4 reuse the Lean code of version 1, which was written before the statement file (see "How the proofs work").

After the 33 statements had been proved, a draft of version 2 was checked paragraph by paragraph against them. Some claims of the paper had no statement: that the product of diagrams is associative and that $\mathcal{B}_n$ and $\mathcal{PB}_n$ are monoids (Section 2), that the sets $\{(\lambda, g, \lambda)\}$ are groups isomorphic to S_r (Section 3), the case $n = 0$ of Theorem 1.1(c), which the abstract uses (now stated at the end of Section 7), that Rees 0-matrix multiplication is associative and that complete mappings of a magma are the transversals of its Cayley table (Sections 2 and 1), and the values $N = 45$ and $N = 105$ (Section 1). For them three further statement files were written on 1 October 2026. Each was reviewed by an independent agent and frozen by its SHA-256 hash before its proofs were accepted:

- the file on associativity, frozen at 04:07 UTC, with hash 626da68a129fb8bbe3181605d131f51d5e5da56b285d92de7ff7594df0183b5e;
- the file on $n = 0$ and the groups, frozen at 09:25 UTC, with hash 7c624d7c6085e0173e1df6fc72ec29c157c15278a5ca901a66b0dc3e08c4a400;
- the file on Rees multiplication, Cayley tables and the two values, frozen at 11:24 UTC, with hash 6d864b914c276adc125b659dedaf43f63167f5bcf5b358394d624d80f6ee8911.

They add ten statements and two auxiliary statements about reachability (one for simple graphs in general, one for stacked pictures), which the proof of associativity uses.

What is proved. The statement file states 33 propositions, and the three supplementary files twelve more; all are listed in Table 2. The file of checks (`Research/Backfill/Paper5/Check.lean`) proves each of these 45 propositions, by a theorem `check_X` whose type is the statement `X` written out in full, for every name `X` of the table. For example, Theorem 1.1(c) is the statement `Thm11c`. In ASCII notation (`alpha` for α, `forall`, `exists`, `->`, `<->`, `<=`, `/\` and `\/` for $\wedge$ and $\vee$, `Nat` for $\mathbb{N}$, `in` for $\in$, `<y, h>` for the element y of a subtype with proof h, and `(. * .)` for the multiplication), it reads, together with the definitions of complete mappings that it uses,

```
def IsCompleteMapping {S : Type*} (op : S -> S -> S) (alpha : S -> S) : Prop :=
  Bijective alpha /\ Bijective (fun x => op x (alpha x))
def HasCM (S : Type*) [Mul S] : Prop :=
  exists alpha : S -> S, IsCompleteMapping (. * .) alpha
def SubsetHasCM {S : Type*} (op : S -> S -> S) (M : Set S) : Prop :=
  exists alpha : M -> M, Bijective alpha /\
    exists h : (forall x : M, op x (alpha x) in M),
      Bijective (fun x : M => (<op x (alpha x), h x> : M))
abbrev Sym (r : Nat) := MulOpposite (Equiv.Perm (Fin r))
def Thm11c : Prop :=
  forall n : Nat, 1 <= n ->
    (SubsetHasCM dmul (B n) <-> SubsetHasCM dmul (PB n)) /\
    (SubsetHasCM dmul (PB n) <-> HasCM (Sym n)) /\
    (HasCM (Sym n) <-> (n = 1 \/ 4 <= n))
```

where `B n` and `PB n` are the sets of encoded diagrams of $\mathcal{B}_n$ and $\mathcal{PB}_n$ and `dmul` is their product (see "Definitions the reader must trust").

How the proofs work. The formal proofs follow the written ones, with the following differences. *Section 4.* Lemma 4.1 is proved directly. The proof of Lemma 4.2 follows the written

In the paper	Lean statement
Theorem 1.1 (a), (b), (c)	`Thm11a`, `Thm11b`, `Thm11c`
$\operatorname{rank}(xy) \le \min(\operatorname{rank} x, \operatorname{rank} y)$; the $\mathcal{J}$-classes are the rank classes; the ranks that occur; the group of units (Section 2)	`RankMul`, `JClassesAreRankClasses`, `RanksOccurring`, `UnitsAreSym`
(R1), also with Mathlib's order of composition	`SymCM`, `SymCMPerm`
(R2)	`ZeroFixed`
the number N of half-diagrams (Section 3)	`CountH`
Proposition 3.1 (a)–(d) and the isomorphism	`Prop31a`, `Prop31b`, `Prop31c`, `Prop31d`, `Prop31iso`
the idempotents e_λ (Section 3)	`IdempotentE`
Lemmas 4.1, 4.2, 4.3, 4.4	`Lemma41`, `Lemma42`, `Lemma43`, `Lemma44`
Remark 4.5 (1), (2), (3)	`Remark45a`, `Remark45b`, `Remark45c`
the fibres (Section 5)	`FibreCount`
Lemmas 5.1, 5.2, 5.3, 5.4	`Lemma51`, `Lemma52`, `Lemma53`, `Lemma54`
the product $e_{I_0}e_{I_1}e_{I_2}e_{I_0}$ (Section 5)	`TriangleProduct`
Lemma 6.1, Proposition 6.2	`Lemma61`, `Prop62`
$\mathcal{PB}_n$ and $\mathcal{B}_n$ are closed under the product, with the identity diagram as identity (a check of the model)	`ModelMonoid`
Supplementary statements	
the product is associative; $\mathcal{PB}_n$ and $\mathcal{B}_n$ are monoids (Section 2)	`PBAssociativity`, `BAssociativity`, `PBMonoidAligned`, `BMonoidAligned`
(auxiliary: reachability in graphs and in stacked pictures)	`ReachableElimination`, `OuterBracketReachability`
the groups $\{(\lambda, g, \lambda)\}$ (Section 3)	`CornerGroupAligned`
Theorem 1.1(c) for every $n \ge 0$ (Section 7)	`AllNClassification`
Rees 0-matrix multiplication is associative (Section 2)	`ReesAssociativity`
complete mappings are the transversals of the Cayley table (Section 1)	`CayleyTransversalIff`
$N = 45$ and $N = 105$ (Section 1)	`HalfCount45`, `HalfCount105`

Table 2: The 33 statements of the statement file, in the namespace `BackfillPaper5.Challenge`, and the 12 statements of the three supplementary files, whose namespaces begin with `CodexPaper5`. The special cases (R3) and (R4) are used only in the proof of Theorem 1.1(c), and are proved inside the proof of `Thm11c`.

proof: the bijectivity of $x \mapsto x \cdot x\bar{f}$ shows that the sandwich entries used are non-zero, the lift is built with the representatives 1 and (1 2) in the order of the written proof, and the lift and its product map are shown to be injective, hence bijective because the sets are finite; that squaring is a bijection of A_r uses that A_r has order 1 or 3. That the maps $\bar{f}$ of Lemmas 4.3 and 4.4 are complete mappings of $\mathcal{M}_2$ is taken from the module `BrauerCMCore` of version 1, in a copy that differs from it only in its namespace and in four uses of two library lemmas whose names have changed. The other parts of these lemmas are new: δ and t exchange the two halves of a splitting of H into two sets of equal size, a Latin square on every non-empty finite set is transported from $\mathbb{Z}_N$, and c is defined on each cycle of ρ from one point of the cycle, shared by all its points and including the step that closes the cycle; then Lemma 4.2 applies. *(R1).* A complete mapping α of S_2 or S_3 would give $\sum_x s(x \cdot x\alpha) = \sum_x s(x) + \sum_x s(x\alpha)$ in $\mathbb{Z}_2$; the right side is 0, since α is a bijection, while the left side is the number $r!/2$ of odd permutations, which is odd. The groups S_0 and S_1 are trivial, the kernel checks a complete mapping of S_4 given by its 24 values, and for $n \ge 4$ a complete mapping of S_n is lifted to S_{n+1} by the construction of Hall and Paige [5, Theorem 1 and the proof of Theorem 2], through the stabiliser of the new point and the transpositions that move it, which form a set of coset representatives that is both

a left and a right transversal; the right action of the paper is matched at the end. *Diagrams and Proposition 3.1.* In the picture of x above y every outer point has at most one partner, which gives the closure of $\mathcal{PB}_n$ and $\mathcal{B}_n$ under the product. The free points, the bijection of the through blocks and the paths of $\Gamma(\lambda, i)$ give the coordinates of Proposition 3.1 and the inequality $\operatorname{rank}(xy) \le \min(\operatorname{rank} x, \operatorname{rank} y)$; within one rank class the rank is kept exactly when the middle halves are compatible, and the product of coordinates and the isomorphism follow. The $\mathcal{J}$-classes, the group of units and the ranks that occur are obtained from explicit diagrams: the diagrams e_λ, products bracketed as in the definition of the $\mathcal{J}$-relation, and permutation diagrams. *Counts.* The half-diagrams with a given free set correspond to the involutions (for $\mathcal{PB}_n$) or the perfect matchings (for $\mathcal{B}_n$) of its complement; the perfect matchings are counted by choosing the partner of one point, which gives the double factorial. Lemmas 5.1 and 6.1 are proved as written above. *Section 5.* Lemma 5.2 follows the paths of its written proof, and the product $e_{I_0} e_{I_1} e_{I_2} e_{I_0}$, bracketed from the left, is computed one factor at a time. Lemma 5.3 uses the points z_t and a matching of the remaining points. Lemma 5.4 pairs the fibres by labels, with the 16 pairs for $(n, r) = (7, 3)$ checked by the kernel, and then pairs the remaining elements of each fibre. *Proposition 6.2.* The isomorphism is built case by case from the three half-diagrams of rank 2 of $\mathcal{PB}_3$ and the coordinates in S_2; the non-existence is the parity argument of the written proof. *Theorem 1.1.* Parts (a) and (b) combine the $\mathcal{J}$-classes, the isomorphism of Proposition 3.1 and the lemmas of Section 4; in the odd case of (a), ρ consists of the 3-cycle of Lemma 5.2 and the 2-cycles of Lemma 5.4. In (c), (R3) is proved for $\mathcal{B}_n$ and $\mathcal{PB}_n$ by summing ranks over the monoid: a complete mapping α and the map $x \mapsto x \cdot x\alpha$ both preserve the rank, so they restrict to complete mappings of the group of units. (R4) is proved by removing the zero of each principal factor and putting the restricted maps of all ranks together, including empty rank classes and the group of units. *Supplementary statements.* Associativity is proved by comparing the pictures of $(xy)z$ and of $x(yz)$ with the picture of all four rows, deleting the inner rows while keeping the reachability between the remaining points; the monoids are then built with the product of diagrams and the identity diagram. The group structure on $\{(\lambda, g, \lambda)\}$ is transported from S_r along the coordinates of Proposition 3.1, and its product and identity are shown to be the product of diagrams and e_λ. Rees 0-matrix multiplication is shown to be associative case by case on the zero entries; the set $\{(x, x\alpha)\}$ is the transversal that corresponds to a complete mapping α; and $N = 45$ and $N = 105$ are instances of the formula for N. Steps of the written proofs that the formal proofs replace, such as the path argument in the proof of Proposition 3.1(b) and the appeal to [1, Corollary 3.4 and Theorem 4.4] in the proof of Theorem 1.1(c), are not formally verified; the results they prove are.

Checks. For every statement of Table 2, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`, or none. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern` or `unsafe`, and no `axiom` declaration, in the package. The Lean package of this paper, 70 modules (the statement file, its tests, the three supplementary statement files, 64 proof modules and the file of checks), was built module by module in two new empty directories, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, and every module was checked with `leanchecker`; all 280 commands ended with exit code 0 and printed no warning (6 October 2026, 16:28–16:44 UTC; the two builds ran side by side). The 68 known-answer tests were compiled and checked in both builds. For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (6 October 2026, 18:56–20:19 UTC). The statement file, its tests, the supplementary statement files, the proofs and the file of checks are in version 1.12.0 of the repository [3], directory `lean/Research/Backfill/Paper5/`; the receipt of the two builds is in `certificates/brauer-complete/lean-package-v2/`, and the logs of the rebuild of the whole project are in `logs/`.

Definitions the reader must trust. From Mathlib the statements use permutations of `Fin r` and their signs, `MulOpposite`, involutive and bijective functions, minimal periods, simple graphs given by a relation and their reachability relation, finite sets, cardinalities (`Set.ncard` and `Nat.card`), binomial coefficients, double factorials, `ZMod 2`, options, sums of types and lists. Mathlib has no complete mappings, Latin squares, Rees matrix semigroups, Green's relations, principal factors or diagram monoids, so the statement file defines them, each in a few lines: `IsCompleteMapping`, `HasCM`, `SubsetHasCM` and `MagmaIso` (complete mappings of an operation, of a type with a multiplication, and of a subset, the products $x \cdot x\alpha$ being required to lie in it, and isomorphisms of operations); `Sym`, `sgn` and `swap12` (S_r with maps acting on the right, the sign with values in $\mathbb{Z}_2$, and $(1\,2)$); `Rees0`, `rees0Mul`, `mulZ2`, `sgnMat`, `sOf` and `delta3` (Rees 0-matrix semigroups, $\mathcal{M}_2$, $s(P)$ and Δ_3); `IsLatin`, `fA`, `fB`, `fC` and `CycleCond` (Latin squares, the maps of Lemmas 4.1, 4.3 and 4.4, and condition $(*)$); `Pt`, `IsPB`, `PB`, `B`, `idD`, `SPt`, `upper`, `lower`, `outer`, `stackEdge`, `stackGraph`, `ptList` and `dmul` (diagrams, the identity diagram and the product); `rank`, `Fam`, `Dset`, `Jr` and `RankOccurs` (ranks and rank classes); `JRel`, `JClass`, `pfMul` and `PFHasCM` (Green's $\mathcal{J}$-relation and principal factors); `Half`, `IsHalf`, `Hr`, `HSub`, `topHalf`, `botHalf`, `pos`, `RepG` and `gCoord` (half-diagrams, top(x), bot(x) and $g(x)$); `GammaEdge`, `gammaGraph`, `Compat`, `RepP`, `pSand`, `sand` and `sandSub` ($\Gamma(\lambda, i)$, compatibility, $p_{\lambda i}$ and P); `eDiag` (e_λ); `r0`, `triPartner` and `triHalf` (R_0 and I_0, I_1, I_2); and `invCount` ($a(k)$). They are checked by the 68 known-answer tests of the statement file, including instances that must come out false; no further known-answer tests were written with the proofs. The supplementary statements use, in addition, Mathlib's semigroups, monoids, groups and multiplicative isomorphisms, graph embeddings and unions of graphs, and the type `Corner` of the diagrams of rank r whose two halves are both λ.

Faithfulness and non-claims. A diagram of $\mathcal{PB}_n$ is represented by the involution x of the set of the $2n$ points whose blocks are the sets $\{p, x(p)\}$; the singletons are its fixed points, and $\mathcal{B}_n$ consists of the involutions without fixed points. The product joins an outer point to the other outer point, if there is one, in its connected component of the picture of x drawn above y (reachability in a simple graph of Mathlib); the statement `ModelMonoid` says that $\mathcal{PB}_n$ and $\mathcal{B}_n$ are closed under this product and that the identity diagram is an identity. That this is the product of Section 2 is a matter of reading its definition, which the known-answer tests support. The 33 statements of the statement file do not use the associativity of the product: Green's $\mathcal{J}$-relation is defined with the bracketing $x = (ay)b$, and "$\mathcal{B}_n$ has a complete mapping" means that some bijection α of $\mathcal{B}_n$ makes $x \mapsto x \cdot x\alpha$ a bijection of $\mathcal{B}_n$. The supplementary statements prove that the product is associative and that $\mathcal{PB}_n$ and $\mathcal{B}_n$ are monoids with this product and the identity diagram. Principal factors are formed from $\mathcal{J}$-classes, always with a new zero, and a principal factor is proper if its elements are not $\mathcal{J}$-related to the identity; Proposition 3.1, Section 5 and Proposition 6.2 use the rank classes J_r. A half-diagram (F, M) is represented by F and the involution of $[n]$ whose 2-cycles are the arcs of M; points are numbered from 0; $g(x)$ and $p_{\lambda i}$ are specified through the positions of points in the free sets and chosen by Lean's choice operator, which matters only where they exist. Compatibility $\lambda \sim i$ is modelled as: every point of $F(\lambda)$ is connected in $\Gamma(\lambda, i)$ to some point of $F(i)$. The group S_r is `MulOpposite (Equiv.Perm (Fin r))`, so that gh means first g, then h; `SymCMPerm` states (R1) also for Mathlib's convention. In Lemma 5.4 a partition into pairs is an involution without fixed points; Remark 4.5(2) is stated for the 2-cycles of a given permutation ρ; Lemma 5.3 is stated for every $r \le n$ with $r \equiv n \pmod 2$, which is more than the paper needs; `IdempotentE` states the coordinates of e_λ, that it is an idempotent and the product rule on the set $\{(\lambda, g, \lambda)\}$, and `CornerGroupAligned` makes this set, with the product of diagrams, a group with identity e_λ that $x \mapsto g(x)$ maps isomorphically onto S_r; in `TriangleProduct` the product is bracketed from the left. The index sets of Section 4 are types in Lean's lowest universe, N is the number of elements of the set of half-diagrams, and $a(k)$ is the number of permutations σ of `Fin k`

with $\sigma^2 = 1$. The formalisation does not describe all complete mappings, does not prove the general Hall–Paige theorem or the general results of [1] and [4], does not identify the groups $\{(\lambda, g, \lambda)\}$ with Green's $\mathcal{H}$-classes, and does not concern the computations of version 1. That the components of $\Gamma(\lambda, i)$ are paths and cycles, which Section 3 uses to describe compatibility, and that the involutions encode the diagrams faithfully, are not separate formal statements: the formal proofs work with reachability in $\Gamma(\lambda, i)$ and with the encoding itself. The agreement of the Lean definitions with those of the paper was checked by the reviewing agents, not by a machine.

Changes in version 2. Apart from this section, Section 8, the abstract, the disclosure and the dates, version 2 differs from version 1 as follows. In Section 3, M in the definition of $\mathrm{top}(x)$ and $\mathrm{bot}(x)$ is now the set of blocks of size 2 in the row; version 1 said "the set of blocks", which for $\mathcal{PB}_n$ includes the singletons, so that $\mathrm{top}(x)$ would not be a half-diagram (the proofs used the intended definition; the error was found by OpenAI Codex, see the disclosure). The results of [1] are now stated in Section 2 only in the special forms that the proofs use: (R1) for the symmetric groups, without the deduction from the Sylow subgroups that version 1 gave; (R3) and (R4) for $\mathcal{B}_n$ and $\mathcal{PB}_n$, and (R4) in one direction. The general lifting lemma (R5) of version 1 is replaced by an attribution in the proof of Lemma 4.2, which now also shows why the sandwich entries it uses are non-zero and why the lift is a bijection; (R6) and (R7) are replaced by attributions after Lemmas 4.1 and 4.3; and Proposition 6.2 is proved by a direct parity argument instead of (R8), that is, [1, Theorem 7.19]. Claims for which we have no formal proof were deleted: Remark 4.5(4) of version 1, which compared Lemma 4.4 with the residual-routing criterion of [1, Definition 7.13 and Theorem 7.14]; the explicit complete mappings of $\mathcal{B}_n$ and $\mathcal{PB}_n$ for $n \in \{1, 4, 5, 6\}$ mentioned in the abstract and the results of all the computations of Section 8; the numbers of elements of the rank-2 factor of $\mathcal{B}_6$ and the rank-3 factor of $\mathcal{B}_7$ (Section 1); the remark after Proposition 6.2 that its conclusion also follows from [1, Theorem 10.1]; that a maximal subgroup in rank r is isomorphic to S_r (Section 2); and the identifications with Green's $\mathcal{L}$-, $\mathcal{R}$- and $\mathcal{H}$-classes in the abstract, Section 1, Table 1 and Sections 3 and 5, which are replaced by the number of half-diagrams and by the groups $\{(\lambda, g, \lambda)\}$. The proofs of Lemmas 5.1 and 6.1 were replaced by the ones that are formalised, which use binomial coefficients modulo 2 and the pairing of permutations with their inverses in place of a 2-adic valuation and a recurrence for $a(k)$. Section 2 now states that the product of diagrams is associative, Section 3 defines $a(k)$ more precisely, and Section 7 states the case $n = 0$ of Theorem 1.1(c), which the abstract uses. The theorems are unchanged, and so are the written proofs apart from those of Lemmas 4.2, 5.1 and 6.1 and Proposition 6.2, and the reference for the $\mathcal{J}$-classes at the end of the proof of Proposition 3.1, which now points to Section 2 instead of to [4].

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents; the agents selected this problem. Claude Code agents found the proofs, wrote the programs and the Lean code of version 1, carried out the literature searches described in the introduction, and drafted the text.

Version 2 was produced from 27 September to 7 October 2026. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote the statement file of Section 9, and an independent Claude Code agent reviewed it against the paper. The same day OpenAI Codex (desktop application; model `gpt-6-astra`) checked version 1 of the paper independently: it re-derived the proofs, checked the certificates of version 1 with programs of its own, and found

the error in the definition of top(x) and bot(x) that version 2 corrects. From 29 September to 1 October 2026 OpenAI Codex (desktop application; its main session used the model `gpt-6-astra` and, from 30 September, `gpt-6.1-sol`) wrote the Lean proofs of the 33 statements; each group of proofs was accepted after two builds from empty directories and a review of the evidence by a Codex agent that had not written it. On 1 October 2026 Codex checked a draft of version 2 paragraph by paragraph against the formal statements, wrote the three supplementary statement files of Section 9, which Codex agents that had not written them reviewed before they were frozen, and proved their statements. On 6 October 2026 Codex (model `gpt-6-astra`) built and checked the assembled Lean package in two new empty directories (Section 9), and a Codex agent that had written none of the proofs reviewed the evidence; for the release Codex also rebuilt the whole Lean project of the repository. Claude Code (model Claude Opus 5.5) drafted the text of version 2. An independent Claude Code agent (model Claude Fable 5.1), which had written neither the paper nor any Lean file, refereed the draft of version 2 on 2 October 2026 (UTC): it compared every statement of Table 2 with the frozen files, re-derived the written proofs, and re-checked the small cases and the three constructions by computations of its own; its corrections were applied. Throughout, "we" refers to this workflow. The author is not a professional mathematician. The author has read version 2 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 2 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

References

[1] J. Araújo, W. Bentz, P. J. Cameron, K. Hendrey and M. Kinyon, Complete mappings of semigroups, arXiv:2608.25092v1, 25 August 2026.

[2] P. J. Cameron, Complete mappings of semigroups, *Peter Cameron's Blog*, 27 August 2026, https://cameroncounts.wordpress.com/2026/08/27/complete-mappings-of-semigroups/.

[3] H. Dong, *ai4math-results*, repository, https://github.com/Horace-Maxwell/ai4math-results, versions 1.4.0 and 1.12.0 (2026); archived at Zenodo under the concept DOI https://doi.org/10.5281/zenodo.22970254.

[4] J. East, J. D. Mitchell, N. Ruškuc and M. Torpey, Congruence lattices of finite diagram monoids, *Adv. Math.* **333** (2018), 931–1003, https://doi.org/10.1016/j.aim.2018.05.016.

[5] M. Hall and L. J. Paige, Complete mappings of finite groups, *Pacific J. Math.* **5** (1955), 541–549, https://doi.org/10.2140/pjm.1955.5.541.

[6] M. Kinyon, Complete mappings for semigroups, slides of a talk at AGTA, University of Naples Federico II, 24 June 2025, https://www.advgrouptheory.com/agta2025/Talks/Kinyon.pdf.

[7] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs* (CPP '20), ACM, 2020, 367–381, https://doi.org/10.1145/3372885.3373824.

[8] The Omega Institute, `trureturing` repository, https://github.com/the-omega-institute/trureturing.